

# A262 – $E = mc^2 = E = m$ : The $c$ -Convention and the Three-Clock Experiment

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## .1 Purpose

**[STIPULATION]** This document addresses two related points: (1)  $E = mc^2$  is in natural units identical to  $E = m$  – the only difference lies in the convention of setting  $c$  as a constant. (2) The three-clock experiment (following Sci. Rep. 2024) shows that all physical quantities – lengths, masses – can be traced back to a single time standard, which metrologically supports the T0 duality  $\tilde{T} \cdot m = 1$ .

## .2 $E = mc^2 = E = m$

**[B]** In natural units ( $c = 1$ ):

$$E = mc^2 = m \times 1^2 = m.$$

This is not an approximation – it is the same equation. The difference between Einstein and T0 lies not in the physics but in the convention.

**[B]** What is  $c$  really?  $c = L/T$  is a ratio of length to time. Einstein's setting  $c = 299\,792\,458$  m/s freezes this ratio and makes the equation  $E = mc^2$  necessary. T0 theory allows  $c$  to be a dynamic ratio – then  $E = m$  is the more fundamental form. Both formulations are physically equivalent; what differs is the ontological status of  $c$ .

**[B] The constant illusion.** Setting  $c = \text{const.}$  is a legitimate normalisation choice, not a constant of nature. With it:

$$E = mc^2 \text{ (in SI units)} \quad \equiv \quad E = m \text{ (in natural units)}.$$

The apparent complexity of  $c^2$  is a consequence of the unit choice, not of the physics. The same holds for  $\hbar$  and  $k_B$  (A085).

**[S] The time dilation paradox.** In the standard view, time is relative, mass is constant. In T0, time is absolute, mass is variable. Both views describe the same measurements; which to prefer remains an ontological choice (A060).

### .3 The Three-Clock Experiment (Sci. Rep. 2024)

**[B]** The experiment (Sci. Rep. 2024, DOI: 10.1038/s41598-024-71907-0) shows: a single time standard suffices as a starting point to define and measure all physical quantities – time intervals, lengths, masses.

**[B] Length measurement from time.** Lengths are determined exclusively from time differences:  $L = c \cdot \Delta t$ . This sets  $c = 1$  (convention) and gives lengths in units of time – exactly what the T0 duality requires.

**[B] Mass measurement from frequencies.** Via the Compton wavelength  $\lambda_C = \hbar/(mc)$  and metrological procedures (Kibble balance), masses can be traced to the time standard:  $m \propto 1/T$  with  $T$  as the Compton period. This is the metrological realisation of  $\tilde{T} \cdot m = 1$ .

**[K] Connection to T0 duality.** The reduction to a single time standard is not new – T0 already achieves this in the units documents (A080/A085). The three-clock experiment is an external, independent confirmation of the metrological consistency of this reduction. It shows that the programme is technically feasible, but presupposes the same convention choice ( $c = 1$ ).

### .4 Reference-Point Revolution

**[S]** T0 theory makes the implicit reference-point shift explicit:

| Epoch     | Reference                               | Consequence    |
|-----------|-----------------------------------------|----------------|
| Antiquity | Earth as centre                         | Epicycles      |
| Newton    | Sun as reference                        | Kepler's laws  |
| Einstein  | $c = \text{const}$ , space as reference | Time dilation  |
| T0        | Time as reference                       | Mass variation |

Every shift is a convention choice – none is “more correct” than another. What counts: which describes the physics more simply.

### .5 Verification Script

- Numerical equivalence  $E = mc^2 = E = m$  for various  $c$  values; length and mass measurement from a single time standard: `python/A_Serie_Skripte/a262_c_konvention.py`

### .6 Open Questions

**The ontological question is not decided.** Which view is preferable – time relative and mass constant, or time absolute and mass variable – is a question of interpretation, not physics. The A-series makes no judgement (A060).

**The three-clock experiment does not cover the full T0 claim.** It confirms the metrological reduction; it does not confirm the geometric derivation of  $\xi$  or the mass formulas.

## **.7 Supersedes**

This document subsumes: Dok. 077 ( $E = mc^2 = E = m$ ,  $c$ -convention, time dilation paradox, reference-point revolution), Dok. 029 (three-clock experiment Sci. Rep. 2024, length and mass measurement from time standard, connection to T0 duality).

## **.8 Use of AI Tools**

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.